2

TOOLS AND EQUIPMENTS

2.1 INTRODUCTION

In older times, most work was accomplished without the use of any tools or equipment. It was very difficult and there was no ease in working. Today there are various kinds of equipment which ease the process of garment construction. It is very important for anyone involved with garment construction to have adequate knowledge of the tools and equipments available so that work is accomplished smoothly and efficiently. Equipments can be categorised according to their use:-

- measuring tools
- drafting tools
- cutting tools
- sewing equipments

But before actually using these tools or equipments let us learn what are tools and equipments, what are the advantage of using them and also how they should be used.

2.2 OBJECTIVES

After going through this lesson, you will be able to do the following:

- list the tools and equipment used in garment construction;
- use these tools and equipment to increase work efficiency, produce more and maintain quality of work.

2.3 WHAT ARE TOOLS AND EQUIPMENT?

Any article used to aid the process of garment construction, from measuring to sewing, is known as tool and equipment. Their use facilitates work and the final product is neat and beautiful.

2.4 ADVANTAGES

1. It ensures quality product.
2. Work is neat and orderly.
3. Saves time and energy.
4. Ensures greater production.

2.5 DRAFTING TOOLS

1. Scale

This is made of a variety of materials like steel, wood or plastic. It is straight and flat and graduated in inches on one side and centimeters on the other side. This is the basic tool used in drafting.

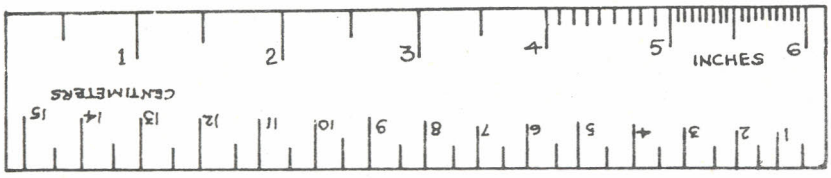


Fig 2.1 Scale

2. L Scale

This, too, is made of metal, wood or plastic. It is shaped like the letter 'L' in English. It has 12" marked on the smaller leg and 24" marked on the longer leg. It is used to simultaneously mark horizontal and vertical lines while drafting. It is also used to mark parallel lines while drafting.

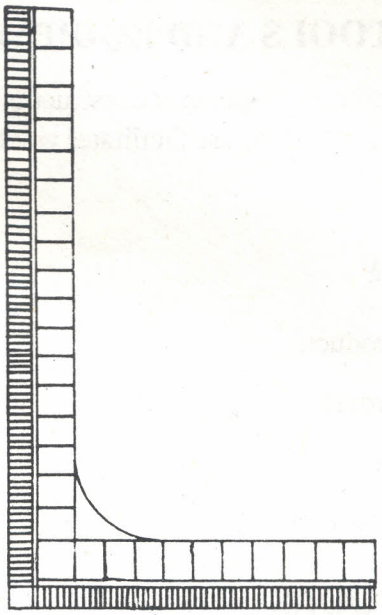


Fig. 2.2 'L' Scale

3. Leg Shaper

This is either wooden or made of plastic. It's length varies from 24" to 30". One end is 3" broad and the other end is 1½" broad. It is marked both in inches as well as in centimeters. The scale has a slight curvature which is used to shape legs of trousers or sleeves for coats, etc.

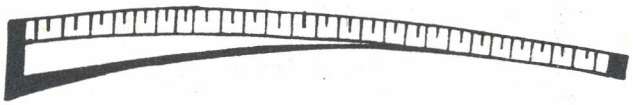


Fig 2.3 Leg shaper

4. Tailor's art curve

This is made either of wood, plastic or iron. This is also 'L' shaped on one side. It has markings of 1/6 on one side and 1/4 on the other.

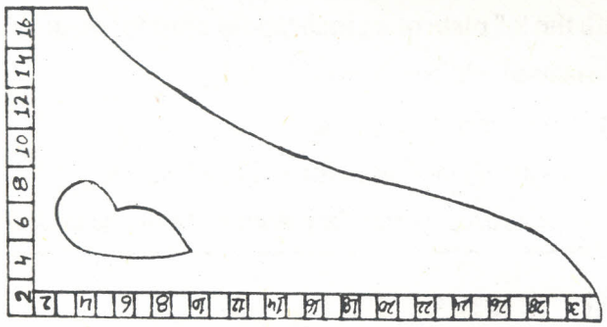


Fig 2.4 Tailor's art curve

5. Shaper

Shapers come is various shapes, some corresponding to the English letter 'S' and 'U'.

These are also known as French curve. These are usually available as a set of 12, but only 3 or 4 are generally used in tailoring. Shapers are made of plastic. They are used in shaping neck lines, armholes, flairs, etc. in fact wherever one needs to give a curved shape.

6. Tracing wheel

It is made of a spiked wheel attached to a wooden handle. It is used to transfer markings from pattern to the fabric or from one layer of fabric to another. It is used for cottons, silks or synthetics but not for woolen fabrics.

7. Inch tape or measuring tape

This is made of cloth or plastic. With a total length of 60", it has a 3" long metal plate at one end and a ½" plate at the other end. The end with the 3" plate is used to measure lengths while the end with the shorter plate is used a measure widths and circular measures. It is marked with inches on one side and centimeters on the other side.

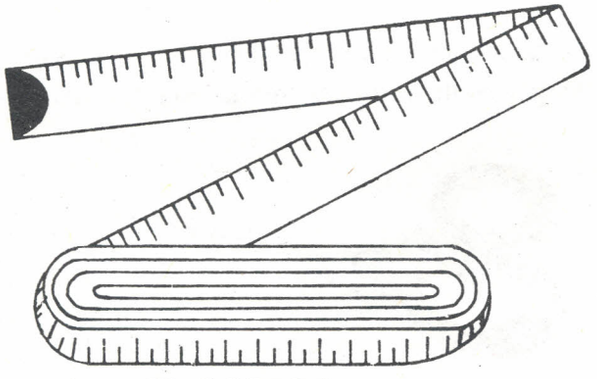


Fig 2.5 Inch tape

INTEXT QUESTIONS 2.1

State whether the following statements are true or false:

a) The end with the ½" plate of an inch tape is used for measuring lengths.

b) The 'L' scale is used while making a drafting.

c) The use of tools helps in saving time.

d) The trouser legs are shaped with the help of L scale.

e) The tracing wheel is used to transfer markings from one layer of fabric to another.

2.6 CUTTING TOOLS

After learning about the drafting tools, let us learn about the cutting tools which are as follows:-

1. Shears

These are scissors which are 10" to 14" long. They are generally used to cut thick fabrics or 6-7 layers of fabric together. Shears are so shaped that they can cut the garment with ease and with as little lifting of the fabric from the surface on which it is being cut. The handles are made so that the thumb fits into one and 2-3 fingers in the other. The lower blade is straight and the upper blade is at an angle.

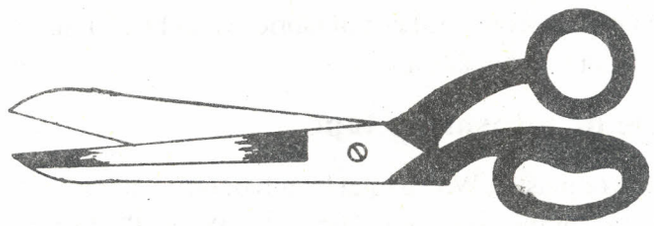


Fig 2.6 Shears

2. Scissors

These are 7" to 10" in length. These are generally used in homes for cutting light material. It is quite light in weight. Both the blades are shaped similarly.

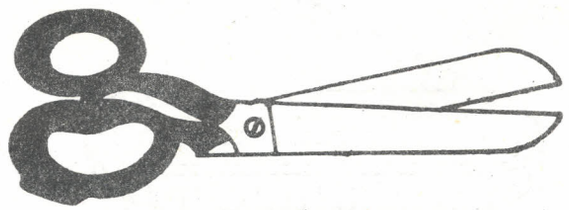


Fig 2.7 Scissors

3. Trimming scissors

These are about 4" to 5" in length. These are used when stitching a garment or for finishing the garment after stitching. It is also used for cutting laces, piping ribbons, strings and embroidery threads.

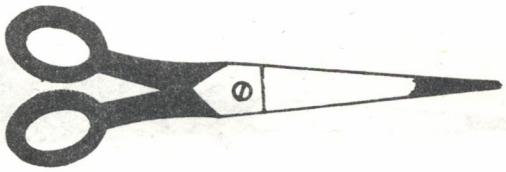


Fig 2.8 Trimming scissors

4. Pinking shears

These are about 8" to 10" in length. They are designed with blades which have notched edges. They are used to finish the raw edges of seams so that the edges do not fray. The edge cut by these shears appears zig-zag. This finish is usually given in ladies and children's garments.

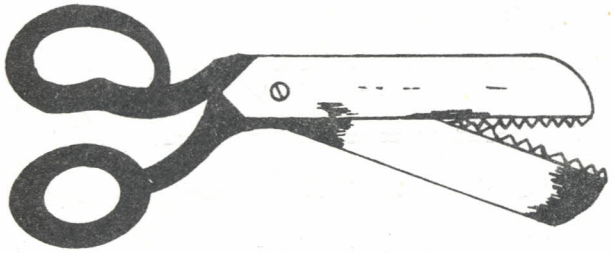


Fig 2.10 Pinking shears

5. Buttonhole scissors

This is used to make buttonholes. These scissors are 4" to 6" in length. They can cut button holes from ½" to 6 points in length. There is a screw which is used for controlling the length of cut for the buttonholes. The cut can be shortened by tightening the screw and lengthed by loosening the screw.

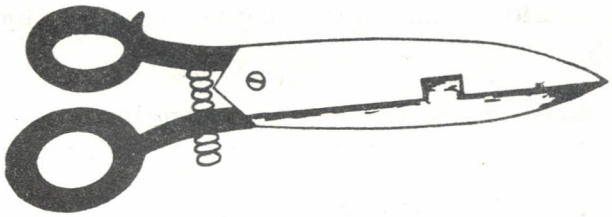


Fig 2.10 Buttonhole scissors

6. Poker

This is also known as a 'hole maker'. It is usually made of metal and is used for making holes. It is rod shaped with one end sharp and pointed and a wooden handle at the other end.

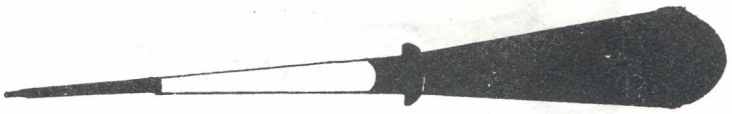


Fig 2.11 Poker

7. Notching scissors

This is also known as a 'truck marker'. It is made of metal. There are notches on the lower edge which are used to mark tucks or pleats.

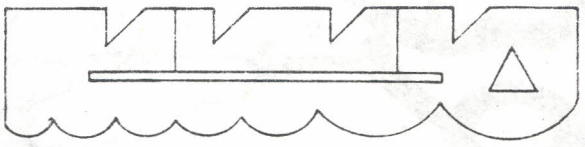


Fig 2.12 Notching scissors

INTEXT QUESTIONS 2.2

Fill in the blanks:

a) Buttonhole scissors are used for making .....

b) Poker is also known as .....

c) Notching scissors are used for marking .....

d) Shears are ..... to ..... inches long and can be used for cutting ..... layers of cloth.

e) Trimming scissors are used for ..... garments.

2.7 SEWING EQUIPMENTS

There are two kinds of needles used in sewing:-

1) Machine needles

2) Hand needles

1. Machine needle

These are also of two kinds -

i) Flat head needles

This type of needle is flat on one side and round on the other. This is generally used in the sewing machines used at home. There are various sizes of needles and they are numbered accordingly. The thicker needles are used for thick or coarse material and finer needles are used for stitching fine material.



Fig 2.13 Flat head needles

ii) Round head needles

This type of needle has a round head. It is generally used in the power, industrial sewing machines.

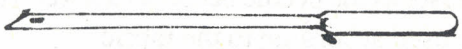


Fig 2.13 Round head needles

Machine needles are available in many brands like Singer, Puf, Usha, etc. You much take care to buy needles of good quality only. All machine needles are numbered from 9 to 24. The number of the needle increases with its thickness. Needles with number 9 are the finest needles.

i) Needle no. 9-11

These are very fine needles. These are used to stitch or embroider on very fine and delicate materials.

ii) Needle no. 12-14

These needles are slightly thicker and larger. These are used on fine calicos, linen, cotton and silk fabrics.

iii) Needle no. 15

These needles are used on fabrics like cambric, organdy, voila, etc.

iv) Needle no.16

These needles can be used to stitch any type of cloth like poplin, long cloth, etc.

v) Needle no.18

These needles are used for stitching jeans, tussar, corduroy and woolen fabric.

vi) Needle no.19

These needles are used for stitching parachute, canvas, etc.

vii) Needle no.21 to 24

These needles are very thick and are generally used in factories. They are used for stitching tarpaulin and unshrunk thick fabrics.

2. Hand needles

Hand needles are of many kinds - some are long, some short, some fine and some thick. The short, thick needles are used for basting and hemming in gents garments. These needles are strong. They are also numbered for convenience. The number increases with the fineness of the needle and decreases as the needle becomes thicker. Hand needles are numbered from 0 to 12. They are used according to the fabric.

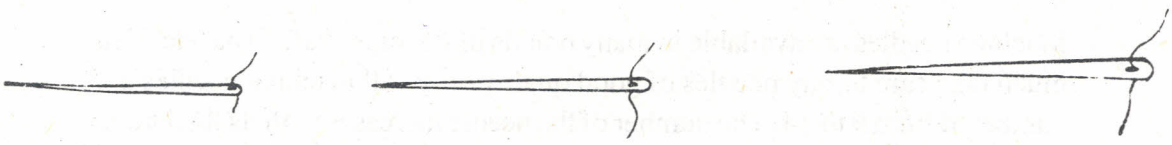


Fig. 2.14 : Hand needles

i) Needle no. 0-1

These are thick needles. They are used to stitch suitcases, for quilting mattresses and for stitching gunny bags, tarpaulins and for leather work.

ii) Needles no. 2-3

These needles are used for upholstery work. Needle no.3 is used to stitch sweaters. It has a long eye. It is also used to make the point of a collar.

iii) Needle no. 4-5

These are used for stitching military uniforms and for basting thick woolen garments. Needle no.5 is used for making buttonholes in gents garments.

iv) Needle no. 6-8

These needles are fine and long and are used for doing hemming on cotton and silk garments.

v) Needle no. 9-10

These needles are very fine and delicate. They are used for embroidery work on chiffon, nylon, terrylene, etc.

vi) Needle no. 11-12

These needles are as fine as hair. The hole is golden coloured. These needle are generally used for fine bead work embroidery.

3. Thimble

This is an aid for protection of fingers during hand sewing. It is made of hard plastic or metal. It comes in two kinds - one shaped like a small glass and the other shaped like a small glass open at both ends. It is usually worn on the middle finger of the right hand so that the needle can be used easily yet keep the finger protected. The outer surface has tiny marks of needle size. Using a thimble lends efficiency to hand stitching.



Fig 2.15: Thimble

2.8 WHAT YOU HAVE LEARNT

In this lesson you have learnt about tools and equipments, their advantages, about drafting tools, about the use of various kinds of cutting tools and about the various kinds of needles used.

ACTIVITY

Describe the following drafting tools-

a) Tailor's art curve

b) Shaper

c) Tracing wheel

2.10 ANSWERS TO INTEXT QUESTIONS

2.1 (a) False (b) True (c) True

(d) False (e) True

2.2 (a) Buttonholes (b) Hole maker

(c) Tucks/pleats (d) 10", 14", 6-7

(e) Finishing